

Collective intelligence in a pandemic

an agent based model of personalised control strategies

Dr. Robert Challen, Dr. Zixuan Liu, Prof. Seth Bullock, Dr. Leon Danon
AI4CI Hub, University of Bristol.

Background

COVID highlighted a lack of models for investigating the effects of complex behavioural interventions and comparing to counterfactuals.

We are developing a biologically inspired stochastic, discrete time, **agent based model** to test the feasibility and design of “collective intelligence” control strategies in a pandemic, such as the deployment of enhanced digital contact tracing apps, versus system wide policy intervention such as lockdowns.

The model can also be used to generate data for validating other statistical models, and training machine learning prediction models.

The simulation is designed to run in parallel on HPC clusters such as the Isambard 3 supercomputer.

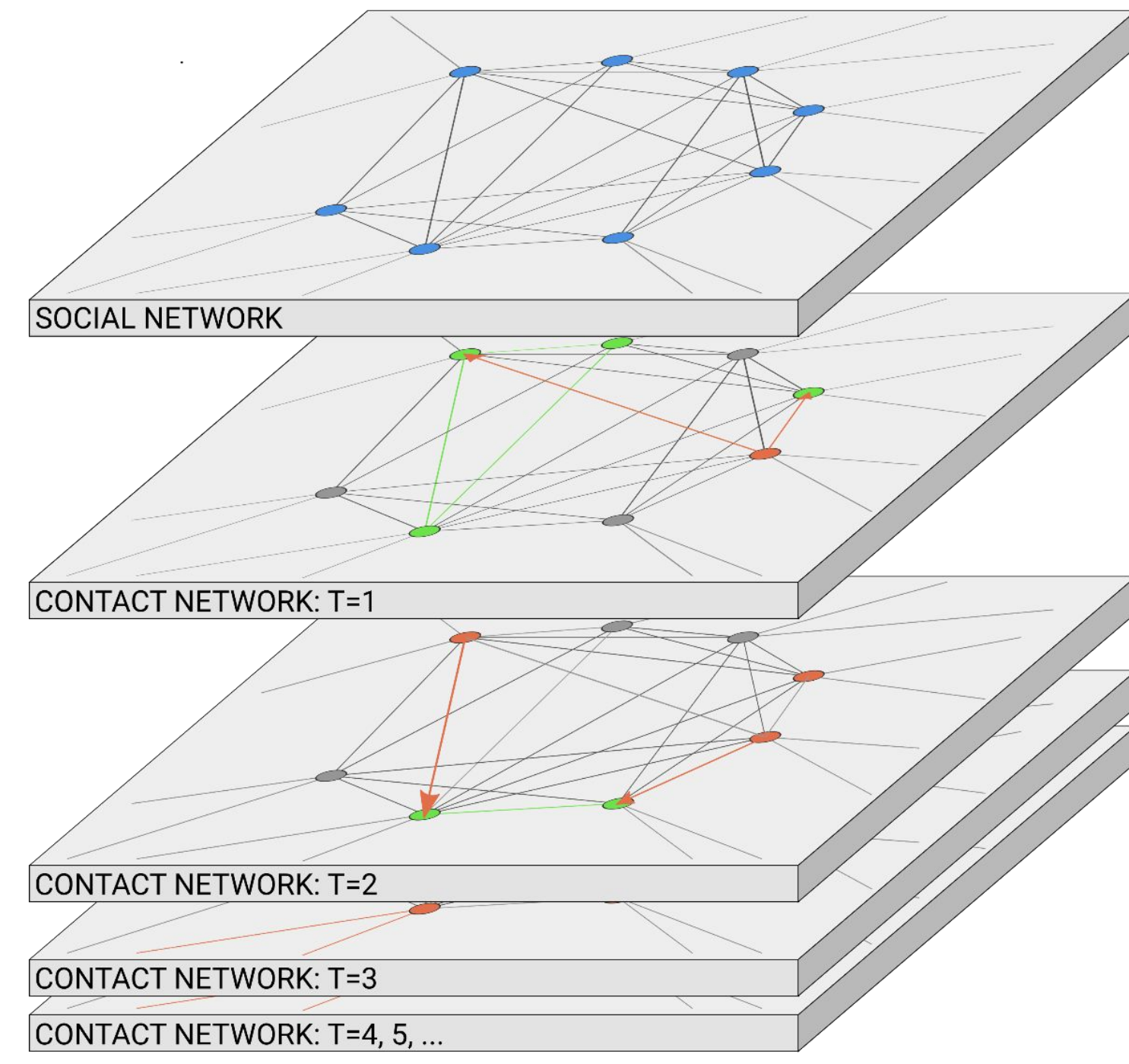


Figure 1 - The conceptual layers of the simulation include a social network, a temporal contact network (green + orange) and within that a network of exposures to infected people (orange arrows). Contacts may be observed by a virtual “digital contact tracing” app. Each node has an in- host viral load model, determining symptoms or test observations, and a state model, which uses the observations to decide agent behaviour.

Behaviour and policy state models

Each agent decides what to do by following one of an extensible set of protocols defined as state models. Driven by factors such as demographics, symptoms, test results, contacts or dynamic “digital contact tracing”, agents may decide to seek testing, reduce their social mobility, or may become non-compliant to guidance. Individuals may change behaviour altogether, for example as a result of central policy.

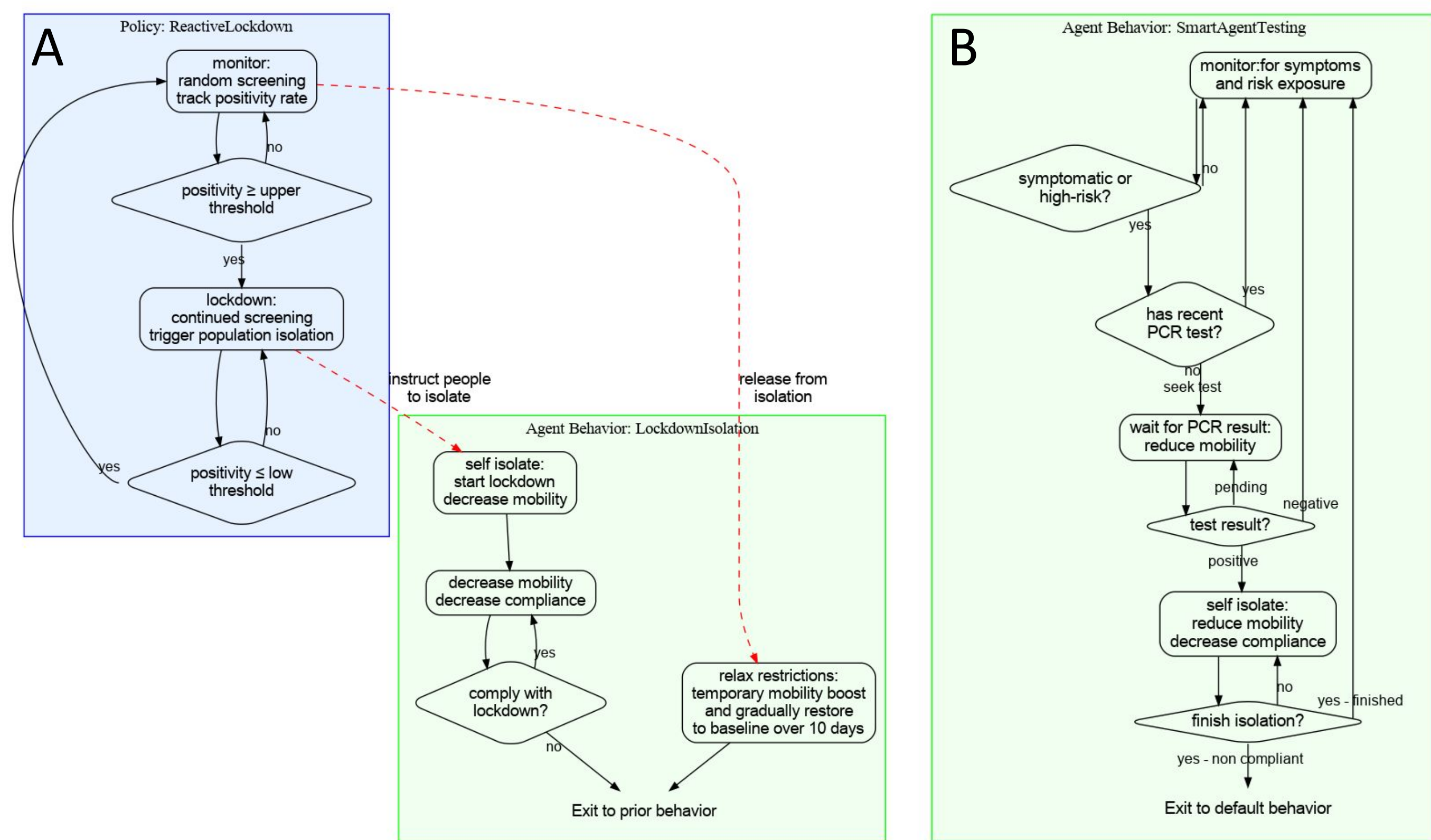


Figure 2 - Example state models. In Panel A a linked policy and individual behaviour model drives simulation wide “lockdown” in response to screening test positivity exceeding a configured threshold. In Panel B, a personal risk assessment, based on symptoms, tests and contact risk, provided by a “smart-agent” drives a decision to seek PCR testing and isolate until the result is known. Other models implemented so far include “symptom-management” - where agents reduce mobility in response to symptoms, “reactive-test” - where agents get a test if they are symptomatic and isolate until they know the result, and a variant of panel B (“smart-agent-lft”) where a quick but less sensitive and specific lateral flow test is used to screen high risk individuals prior to a definitive PCR test.

References

- (1) M. J. Keeling and K. T. D. Eames, ‘Networks and epidemic models’, *Journal of The Royal Society Interface*, vol. 2, no. 4, pp. 295–307, Jun. 2005, doi: 10.1098/rsif.2005.0051.
- (2) L. Danon et al., ‘Networks and the Epidemiology of Infectious Disease’, *Interdisciplinary Perspectives on Infectious Diseases*, vol. 2011, no. 1, p. 284909, 2011, doi: 10.1155/2011/284909.
- (3) O. Puhach, B. Meyer, and I. Eckerle, ‘SARS-CoV-2 viral load and shedding kinetics’, *Nat Rev Microbiol*, vol. 21, no. 3, pp. 147–161, Mar. 2023, doi: 10.1038/s41579-022-00822-w.
- (4) M. J. Plank, et al. ‘Potential reduction in transmission of COVID-19 by digital contact tracing systems: a modelling study’, *Mathematical Medicine and Biology: A Journal of the IMA*, vol. 39, no. 2, pp. 156–168, Jun. 2022, doi: 10.1093/imammb/dqac002.

Social network and demographics

Network topology and demographics influence the spread of disease. Factors such as age, influence the outcome of infection, and can be specified in the simulation. There are extensible options for the algorithm that generates the social network (e.g. Erdos-Reyni, Watts-Strogatz, or Barabasi-Albert).

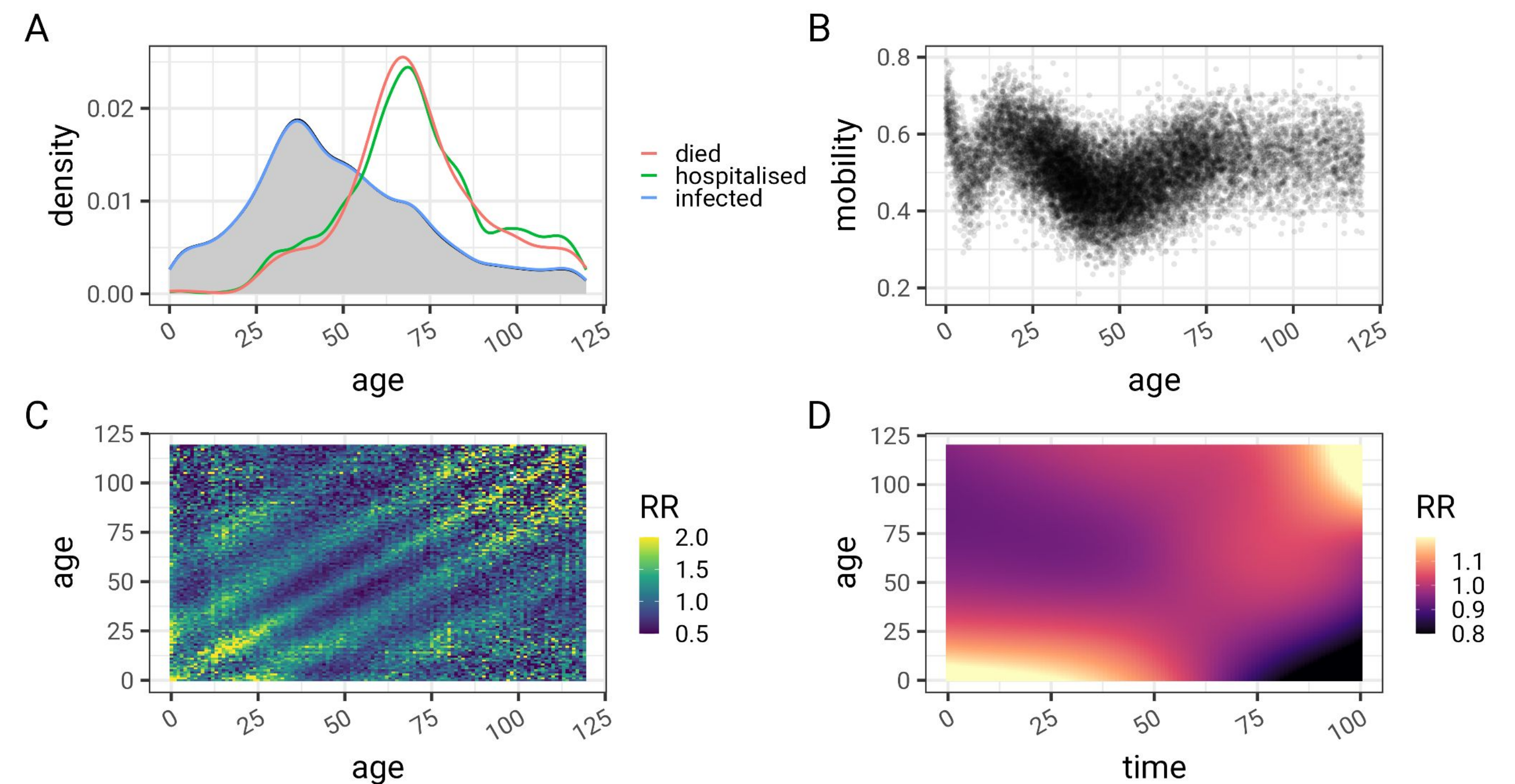


Figure 3 - In panel A, a simulation is configured with a population age distribution (grey). At the end of the simulation risk of infection was uniform across the population (blue), elderly people are at higher risk of hospitalisation (green) and death (orange). In Panel B each of the agents has a different initial social mobility depending on age. Social network links are determined by age difference, and result in contact patterns as seen in in Panel C. Although in panel A infection looks uniform across ages, in panel D we see this evolves during this simulation; the relative risk of infection started highest in the young, before moving to older age groups as a result of panels B and C.

Comparative analyses of outbreaks

Using the simulation different system policies and individual behaviour can be analysed controlling for other factors, e.g. the effect of lockdowns in the presence of symptomatic test and isolation (fig 4), or the final outcomes of different individual behaviour strategies (fig 5).

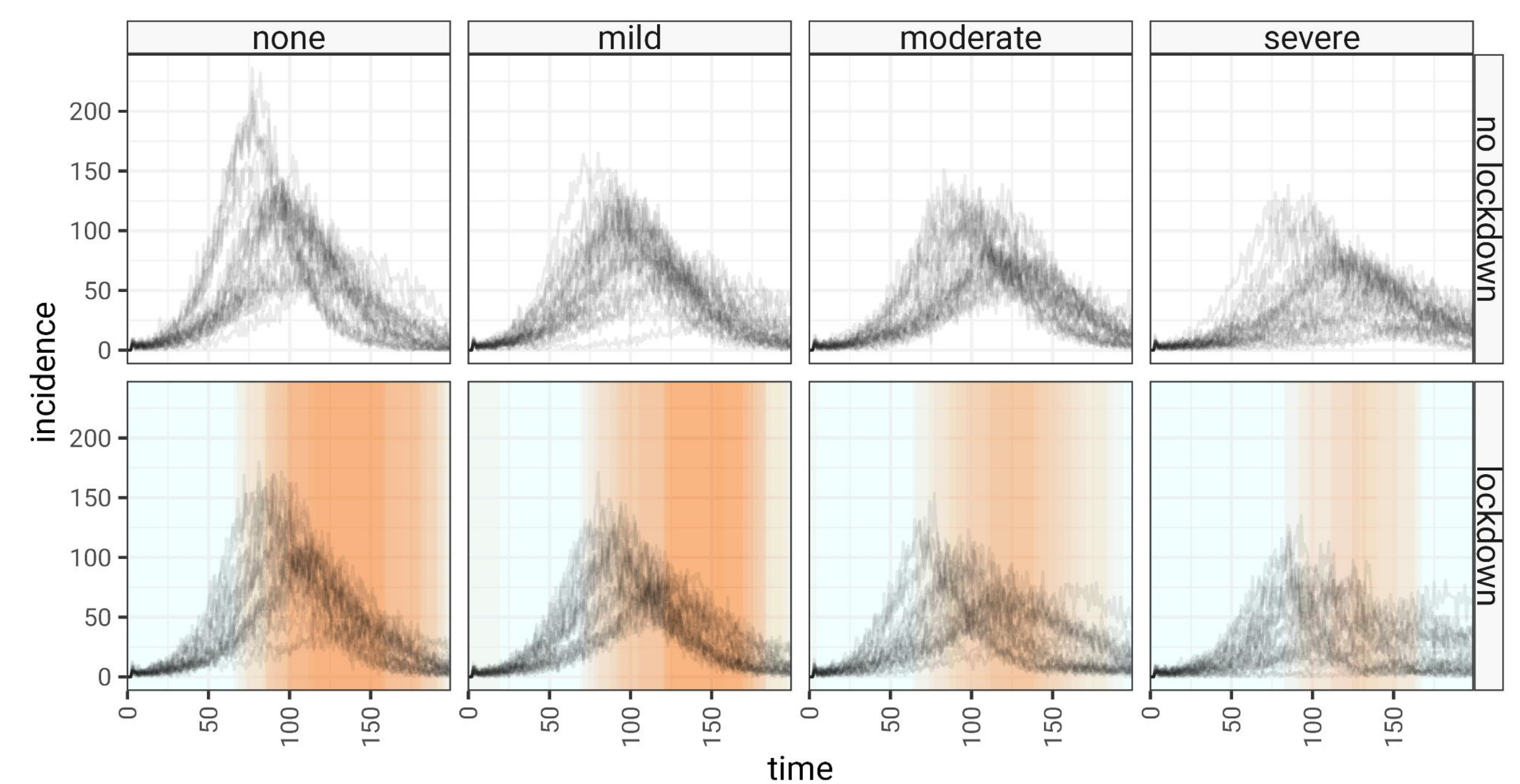


Figure 4 - In a scenario with reactive test and isolation, varying by degree of isolation (columns), lockdowns are triggered by high screening test positivity (orange). Against the background of reactive testing, lockdowns only achieve modest additional gains, and only in scenarios with stringent social mobility reductions during self isolation.

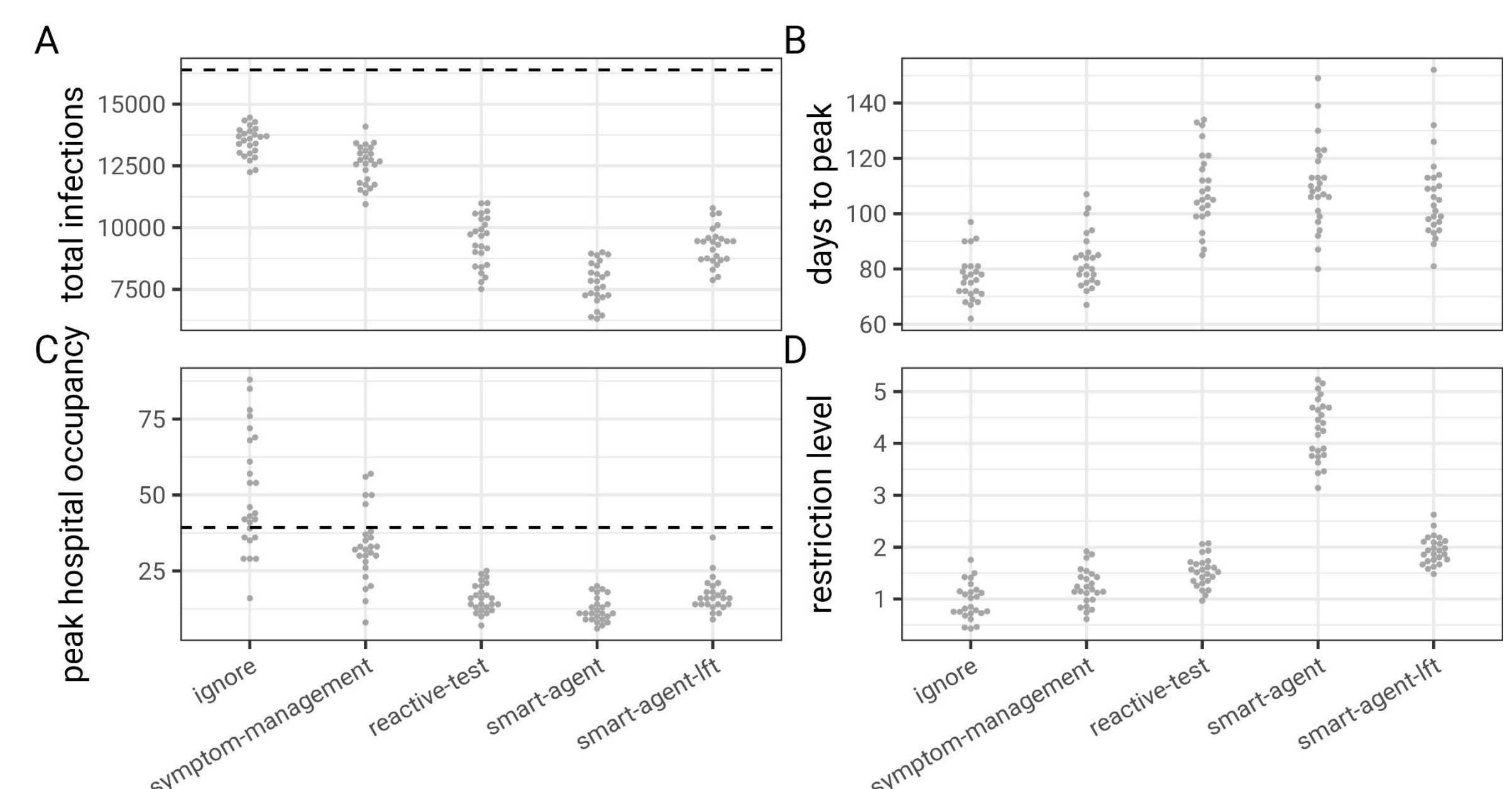


Figure 5 - Outcomes of different behaviour protocols, including no change in response to infection (“ignore”) and other behaviours as described in fig 2. The final size and peak hospitalisations are lower when smart agents are used (panels A&C), and peak timing delayed (panel B). Social restriction is highest for smart agent (panel D) but improved if rapid diagnostic tests (LFT) form part of the protocol.